



Paycom Integration

Productive Hours Calculation In AgriErp

Product Specifications Document

Version 1.0

Table of contents:

Confidentiality	3
Document Properties	3
Review & Distribution List	3
Revision History	4
1.1 Purpose	4
1.2 Scope	4
1.3 Audience	4
1.4 Functional Assumptions	4
Executive Summary	4
Problem Statement:	5
Solution Overview:	5
Pre-requisites	6
Workflow	6
Phase 2 User Journey Workflow: Link	7
Mocks (For Prototype Click Here)	8
There are three Major Roles in a Work order	12
App Users	13
D365 F&SCM	14
Web	15
Resource and Machine Hour Distribution Logic	15
Overlapping Entry Validation and Recalculation Prompt	16
Mobile	16
WorkOrder User Journey:	17
Business Use Cases	19
Validations	20

Confidentiality

The information contained in this document is confidential and proprietary to Folio3. Folio3 submits this document with the understanding that it will be held in strict confidence and will not be disclosed, duplicated, or used, in whole or in part, for any purpose other than the evaluation of Folio3 qualifications, without the prior written consent of Folio3.

For Freedom of Information legislation, Folio 3 classifies this document as “Commercial-in-Confidence” and is governed as per the terms specified in the NDA between Folio3.

Disclaimer

Folio3 shall not be liable for any losses, damages, costs, or expenses arising out of errors in the preparation of this document, due to the provision to it of false, misleading, or incomplete information or documentation or due to any acts or omissions of any other person. The Client indemnifies Folio3 from any liability it may have to the Client or any third party because of any information supplied to Folio3 by the Client or any of its agents, where such information and documentation is false, misleading or incomplete in any material respect

Document Properties

Item	Name
Document Title	Paycom Integration
Author	Folio3
Created By	Wania Aslam
Date	Jun 9th , 2026

Review & Distribution List

Organization	Name	Designation
Folio3	Muhammad Fahad	Head of ERP/CRM Practice
Folio3	Raza Rabbani	Software/Technical Architect
Folio3	Muzzamil Zahid	Solution Architect
Folio3	Owais Shakeel	Senior Project Manager
Folio3	Hassan Ur Rehman	Senior Product Manager
Folio3	Shuraim Ahmed	Senior Functional Consultant
Folio3	Wania Aslam	Product manager

Revision History

Date	Release by Name	Version	Revision Items
Jun 9th, 2026	Wania	1.0	

1.1 Purpose

The purpose of this document is to provide an end-to-end process flow and requirement for the Resource Hours from Farm App (Web & Mobile App) solution and D365 Finance & Supply Chain Management (F&SCM). This document should be used as guidance during detailed design, build, and other project-related activities. This document aims to provide a solution baseline. This document is intended to define and lock the baseline solution approach from the process flow perspective.

1.2 Scope

This document is the primary part of the Farm App design and it is dedicated to the Farm App (Mobile & Web App). This document should be read in conjunction with the DevOps requirements to provide the full understanding of the entire Farm App solution designed for D365 F&SCM integration.

1.3 Audience

The primary audiences for this document are Folio3 project executives, management, and wider implementation teams.

1.4 Functional Assumptions

These workflows will be used as a primary design artifact to confirm existing (delivered)

Executive Summary

This Document is created to fill up the gap between teams in the understanding of the Work Order Management module, its functionality, and the stages the data goes through during the entire Farm App processing.

Problem Statement:

Currently, payroll and labor costing are handled manually through Paycom reports and Excel. The reports are downloaded, adjusted, mapped, and prepared for D365 upload manually. G/L mapping is also done manually.

Solution Overview:

The proposed solution is to introduce ERP and AgriERP into the labor costing process through two separate integrations: Paycom to ERP and AgriERP to ERP.

Paycom will remain the source for actual payroll processing. Employees will continue to punch in and punch out in Paycom, and Paycom will provide actual employee hours, payroll rates, payroll amount, taxes, fees, and compliance-related payroll data. This payroll cost will be posted to ERP. In ERP, indirect labor cost will be expensed directly to the G/L with the cost center, while direct labor cost will be parked in a payroll parking/clearing or absorption account.

AgriERP will be used to capture productive Work Order hours. Farm resources will start and end their Work Order jobs in AgriERP, and the system will capture productive hours against the relevant project, field, season/crop year, activity/operation, and resource. These productive hours will be integrated from AgriERP to ERP and posted as project costing based on the standard/estimated AgriERP resource hourly rate.

Once both Paycom payroll cost and AgriERP productive project costing are available in ERP, the system will calculate the variance by comparing Paycom total hours vs AgriERP productive hours and Paycom actual payroll rate vs AgriERP estimated resource rate. This variance may represent non-productive hours, missed productive Work Order hours, or the rate difference between actual payroll cost and standard AgriERP costing.

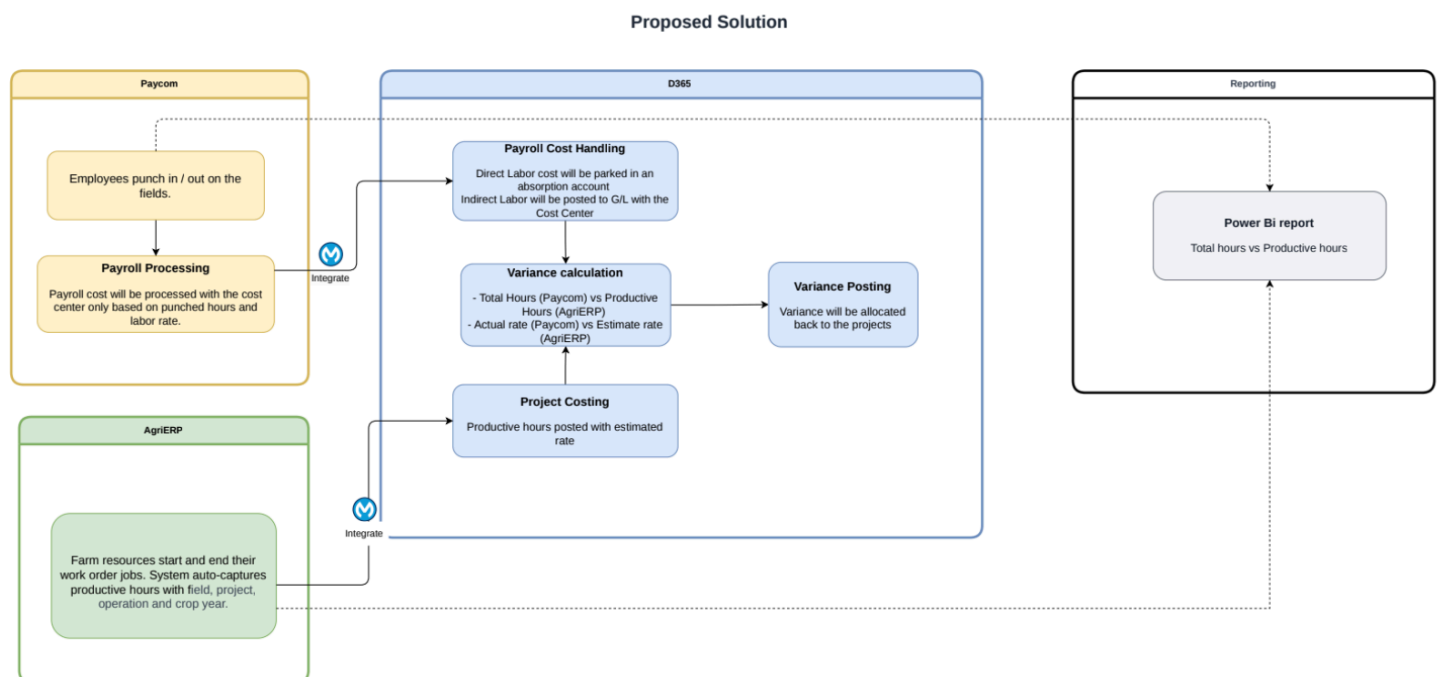
The variance will then be reviewed and allocated back to the projects as required. A Power BI reporting view will also be required to compare Paycom hours with AgriERP productive hours, so the Manager can identify productive hours, non-productive hours and payroll/project costing variances.

Phase 1 Document overview: [link](#)

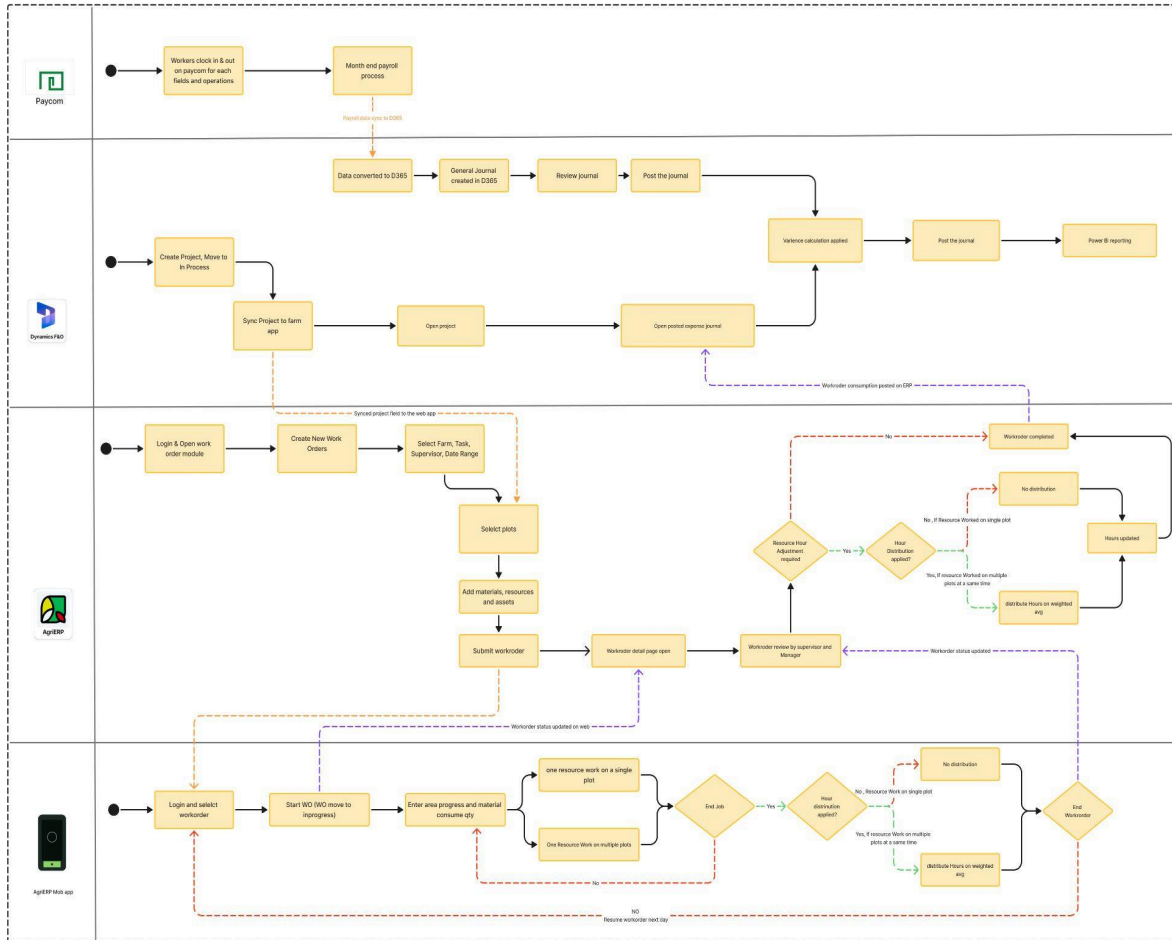
Pre-requisites

- The labour/supervisor needs to add all productive resource hours in AgriERP against the correct Work Orders.
 - Supervisor will be responsible for reviewing and managing the productive hours i.e missed productive hours, productive hours greater than paycom shift in / shift out time etc before posting
- Paycom employees should be mapped with the corresponding AgriERP resources.

Workflow



Phase 2 User Journey Workflow: [Link](#)



Mocks (For Prototype [Click Here](#))

agrierp

Maps | Planning | Work Orders | Template Management | Messaging | Attendance

Grower: testgrower

Crop Cycle: 2022 y...

CP

Home > Work Orders

Work Orders

Inspection Work Orders

Harvest Work Orders

Point of Interests

Create New Work Orders

All

Queue

Draft

To Do

In Progress

Review

Done

Filter

WO Sequence No.	WO Name	Priority	WO Start Date	WO End Date	Site	Plots	Progress	Task
WO-000001	test desc-08/11/2023-1	Medium	08/11/2023	18/11/2023	testgrower_test_site_t...	100303	0.00% Completed	test desc

Page Size100

Page 1 of 1 - Showing 1 - 1 of 1 records

agrierp

Maps | Planning | Work Orders | Template Management | Messaging | Attendance

Grower: testgrower

Crop Cycle: 2022 y...

CP

Home > Work Orders

Work Orders

Inspection Work Orders

Harvest Work Orders

Point of Interests

Create New Work Orders

All

Queue

Draft

To Do

In Progress

Review

Done

Filter

WO-000001	test desc-08/11/2023-1	Medium	08/11/2023	18/11/2023	testgrower_test_site_t...	100303	0
-----------	------------------------	--------	------------	------------	---------------------------	--------	---

Page Size100

Page 1 of 1 - Showing 1 - 1 of 1 records

Summary

Submit

General

Plots

Materials

Resources

agrierp

Maps | Planning | Work Orders | Template Management | Messaging | Attendance

Grower: testgrower | Crop Cycle: 2022 y...

Home > Work Orders

Work Orders

Inspection Work Orders | Harvest Work Orders | Point of Interest

Create New Work Orders

All | Queue | Draft | To Do | In Progress | Review | Done | Filter

WO Sequence #	WO Name	Priority	WO Start Date	WO End Date	Site	Plots	Progress	Task	Status	Created By
WO-000001	test desc-08/11/2...	Medium	08/11/2023	18/11/2023	testgrower_test_si...	100303	0.00% Completed	test desc	Draft	Crop Planner

Page Size: 100 | Page 1 of 1 - Showing 1 - 1 of 1 records

General

Title
test desc-08/11/2023-1

Location
testgrower

Start Date
08/11/2023

End Date
18/11/2023

Task/Job Task
test desc

Status
Draft

Plots

1 Plots Selected
100303 (J00380) PLANNED

Materials

Recyclable Bag 4lb (1113) 1 PCS

Seed - Clearwater Russet (0034) 1 BAG

Resources

Marty Horst Supervisor

agrierp

Maps | Planning | Work Orders | Template Management | Messaging | Attendance

Grower: testgrower | Crop Cycle: 2022 y...

Home > Work Orders > WO-000001

< Work Order: WO-000001 | test desc-08/11/2023-1 Draft

Start Date
08/11/2023

End Date
18/11/2023

Progress
0.00% Completed

Supervisor
Marty Horst [View Spent Hours](#)

Site
testgrower_test_site_test_site

Work Order Type
Planned

Task Type
Planned

Task
test desc

Crop Phase
testing

Machines
0 Machines, 6 Resources

Material Template
N/A

Created By
Crop Planner

Inspection Template
N/A

Responsible
N/A

Notes
N/A

Plots

Seq.#	Plot	Customer Name	Progress	Crop Variety	Total Area - ac	Operational Area - ac	Spent Hours	Attachments
PL-1	100303 (J00380)...	New Concepts Furniture	0 / 0 Beds Completed	Bicycle GENERIC ...	2,200	2,200	0 / 0 Hours Logged	View Attachment(s)

Materials

Name	Rate	Unit	Consumed Qty / Total Qty	View Details
Recyclable Bag 4lb (1113)	1	PCS	0 / 2,200 L	>

Home > Work Orders > WO-000001

< Work Order: WO-000001 | test desc-08/11/2023-1 Draft

Seed - Clearwater Russet (0034)

1

BAG

0 / 2,200 L

>

Resources

Resource Name	Resource Type	Company	Tracking Log	Spent Hours
Resource Group Name: Machine Operator Group				
Ester	Machine Operator		N/A	N/A
Jim	Machine Operator		N/A	N/A
Katherine Hull	Machine Operator		N/A	N/A
Lina Townsend	Machine Operator		N/A	N/A
Robin	Machine Operator		N/A	N/A
Terry Dodds	Machine Operator		N/A	N/A

Machines

Name	Machine Type	Spent Hours	Status	Prestart Checklist
No Record Found				

Plot Progress

Plot	Start Date & Time	End Date & Time	Plan
------	-------------------	-----------------	------

Home > Work Orders > WO-000001

< Work Order: WO-000001 | test desc-08/11/2023-1 Draft

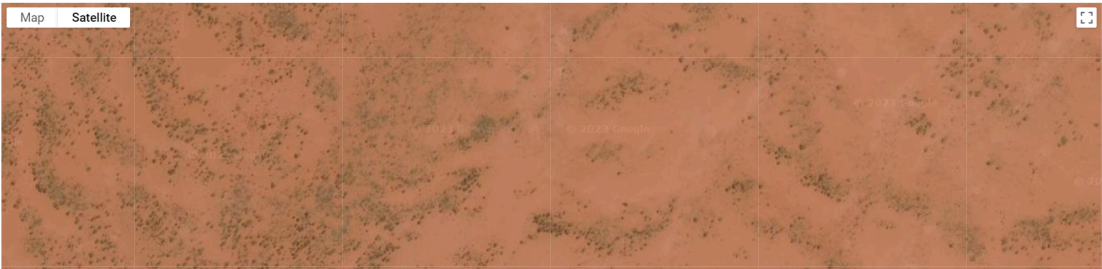
Submit

Plot Progress

Plot	Start Date & Time	End Date & Time	Plan
100303			48362

Plot Map

No coordinates found to load map test_plot.



Creating a new Work Order

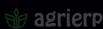
Home > Work Orders

Work Orders

Work Orders | Inspection Work Orders | Harvest Work Orders | Point of Interest

All | Queue | Draft | To Do | In Progress | Planned | General | Filter

WO Sequence No.	WO Name	Priority	WO Start Date	WO End Date	Site	Plots	Progress	Task
WO-000001	test desc-08/11/2023-1	Medium	08/11/2023	18/11/2023	testgrower_test_site_1...	100303	0.00% Completed	test desc



Maps | Planning | W...

Home > Work Orders

Work Orders

Work Orders | Inspection Work Orders | H...

WO Sequence No.	WO Name
WO-000001	test desc-08/11/2023-1

Page Size 100 | Page 1 of 1 - Showing 1 - 1 of 1 records
https://demo2.folio3.com:4444/#block-sum

Create New Work Order Planned

Save As | Submit | X

Work Order Name*
(Task)-12/11/2023

WO Start Date*
12/11/2023

WO End Date*
22/11/2023

Priority*
Medium

Site*
[Dropdown]

Task Type*
Planned

Task*
[Dropdown]

Supervisor*
[Dropdown]

Responsible
[Dropdown]

Inspection Templates*
[Dropdown]

Material Template 
[Dropdown]

Notes
[Text Area]

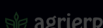
Select Plot

Task Date Range
Select Range

Select	Plot	Customer Name	Plot Area	Job Task Area	Parent
☒	100401 (JOB101) ...	New Concepts Furniture	1250.00 ac	1250.00 ac	

Summary

General | Plots | Resources



Maps | Planning | W...

Home > Work Orders

Work Orders

Work Orders | Inspection Work Orders | H...

WO Sequence No.	WO Name
WO-000001	test desc-08/11/2023-1

Page Size 100 | Page 1 of 1 - Showing 1 - 1 of 1 records
https://demo2.folio3.com:4444/#block-sum

Create New Work Order Planned

Save As | Submit | X

Select	Plot	Customer Name	Plot Area	Job Task Area	Parent
☒	100401 (JOB101) ...	New Concepts Furniture	1250.00 ac	1250.00 ac	

Select Materials

Material	Active Ingredients	Rate	Unit	Unit Cost	Cost Per
Recyclable Bag 4lb (1...	N/A	1	PCS	0	0
Seed - Clearwater Rus...	N/A	1	BAG	0	0

Select Resources

☒ By Resources ☐ By Groups

Resource Name	Resource Type	Company Name	WHS/SDC	Action
Marty Horst	Machine Opera...			

Select Machines

Machine Name	Machine Type	Resource Group	Action
Seed Drills - Tractor	Machine	Machine Group	

Summary

General | Plots | Materials | Resources | Machines

There are three Major Roles in a Work order

App Access	Roles	Description
Web/Mobile	Manager	The manager(office staff) usually uses the farm app but he can access mobile as well. The manager is responsible for • Creating work orders • Review field staff hours and adjustments • Creating a Template for Inspection and Materials
Mobile	Supervisor	The Supervisor is a Field Staff Member usually a permanent worker, Supervisors mainly use the mobile app. Supervisors are responsible for • Open or Start Work Order • Can Start Job for any other farm users • Can assign & change responsibilities • Can Add and remove Farm resources from the Work Order. • Can adjust progress if required

Mobile	Responsible	A single individual could be a machine operator or farmhand, designated as the responsible person for entering Progress Updates / Completions. The Responsible Person is a human resource. Only one per work order. if the Responsible Person clocks off the work order, it may remain open for other workers who may be logged in, however, no progress updates will be requested or asked from the remaining workers on the work order It is also not necessary for the responsible worker to be present/clocked in to commence the work order; any human resource scheduled on that order can commence the work order
--------	-------------	---

App Users

- Managers (Back office staff)
- Supervisor
- Responsible
- Machine Operators
- Farm Hand

D365 F&SCM

The required project activity information from D365 F&SCM will be synced to the Farm app and will be available in the Planning Screen of the AgriERP Web app. Each synced activity will be connected with its related Project ID and activity ID. If any work does not exist as a predefined activity, an adhoc activity will be used in AgriERP to capture and execute that work.

After submitting a Work Order in AgriERP and moving it to the “In Progress” state, the Work Order number will be synced back to D365 F&SCM against the relevant project activity ID for tracking and reference. If the activity is adhoc, the Work Order title will be synced to D365 F&SCM as the activity description.

Paycom will remain the source for actual employee attendance and payroll processing. Employees will continue to punch in and punch out in Paycom, and Paycom will provide the actual payroll hours, payroll rates, payroll amount, taxes, fees, and other payroll-related information to ERP.

AgriERP will be used to capture productive Work Order hours against the relevant project, field, season/crop year, activity/operation, and resource. These productive resource hours will be synced from AgriERP to D365 F&SCM and posted as project costing through an expense journal based on the standard/estimated AgriERP resource hourly rate.

Once the job execution is approved by the Farm Manager, AgriERP will sync the approved consumption details to D365 F&SCM. For material consumption, an item journal will be created and posted in D365 F&SCM. For resource hours and machine hours, an expense journal will be created and posted in D365 F&SCM to record the consumed hours and related cost against the project.

When both Paycom payroll cost and AgriERP productive project costing are available in ERP, the variance can be calculated by comparing Paycom actual payroll hours/cost with AgriERP productive Work Order hours/cost. This variance may represent non-productive hours, missed productive Work Order hours, or the difference between actual payroll cost and estimated AgriERP project costing.

When AgriERP attempts to sync consumption, D365 F&SCM will return a success message with the journal number for reference and tracking if the journal is posted successfully. If the posting fails, D365 F&SCM will return a failure message for correction and retry.

Resource Hour expense journal Document: [link](#)

Variance Calculation Document: [link](#)

Web

Managers can create Work orders against tasks. All work orders sync to mobile and upsync from mobile to the web app and then from the web app to D365.

The manager can adjust resource and machine hours and Materials consumption from the work order details page and approve that after work orders are posted to the D365 .

Resource and Machine Hour Distribution Logic

The system will calculate resource and machine hours based on the actual start and end time of the job. When the same resource and machine are working on a single plot, the complete job duration will be assigned to that plot.

If the same resource and machine are working on multiple plots during overlapping time, the system will distribute the available hours based on plot acreage. The system will calculate the total active acres for all plots running during the same time segment and then allocate the hours to each plot based on its acre ratio.

The same logic will also apply when the same resource and machine are used across multiple work orders. The system will not calculate hours separately for each work order without checking overlap. Instead, it will identify all active plots across all work orders for the same resource and machine combination and distribute the overlapping hours using acre-weighted allocation.

For adjustment scenarios, if the start time, end time, plot, resource, or machine is updated, the system will recalculate the hours for the updated record and all impacted overlapping records. The system will compare the previously calculated hours with the newly calculated hours and determine the adjustment impact.

The adjustment impact will be calculated as the difference between the hours after update and the hours before update. If the difference is positive, additional hours will be added. If the difference is negative, hours will be reduced.

This logic ensures that resource and machine hours are not double-counted and that the total distributed hours remain equal to the actual working duration for the same resource and machine.

After posting system will not allow distribution

Formula:

Weighted Hours = Overlap Hours × Plot Acres / Total Active Acres

Use cases:[Weighted Hours](#)

Overlapping Entry Validation and Recalculation Prompt

When a user creates a new job entry with the same resource, machine, plot, and overlapping time, the system will validate whether an existing entry already exists for the same combination.

If an overlapping entry exists within the same work order, the system will display a confirmation prompt to the user before proceeding. The prompt should inform the user that an existing entry is already available for the same overlapping time and ask whether they want to distribute the existing entry and continue with the new entry.

If the user confirms, the system will distribute the existing entry within that work order and another work order and create the new entry. After the new entry is created, the system must recheck all impacted records for the same resource across other plots and other work orders. If any overlapping time exists, the system will recalculate the resource and machine hours using acre-weighted distribution.

The recalculation should not be limited to the current work order only. The system must also include other active plots and work orders where the same resource and machine are used during the same overlapping time period.

The system should clearly notify the user that confirming the action may impact hour distribution for other overlapping plots or work orders.

Prompt:

An overlapping job entry already exists for this resource and machine.

If you continue, the existing entry in this work order will be distributed and the system will recalculate hour distribution for all impacted overlapping plots and work orders.

Do you want to continue?

Mobile

From the Mobile app Supervisor/Responsible person starts WO, and after its move to **In Progress** every other user Operator/Farm/Crew can start/end their jobs Independently. The app will record resource hours and progress and posts to the farm app, any resource can enter the progress against their assigned job by acres which upsyns to the farm web app.

WorkOrder User Journey:

Personas:

1. Steve: Planning Manager | D365
2. David Luke: Farm Operation Manager | Web App user
3. Martin Waltcher: Machine Operator

Business Central

Steve | Planning Manager

- Steve sign in to the D365 F&O with his credentials
- He goes to the operations (navigation: AgriERP Management ➡ Setup ➡ Farm Operations)
- He creates all required operations, if not available already.
- He then goes to the Project management and accounting ➡ Setup ➡ Projects ➡ Work breakdown structure templates
- He defines the crop, variety & irrigation method, creates a template of WBS using operations, if not created already. Optionally, he can add budgeted / planning lines under the operations.
- He then goes to Project management and accounting ➡ Projects ➡ All Projects ➡ New.
- After reviewing the project, he change the project stage to inprocess.
- Project syncs to the Farm web app and the project are available to make Work Orders.

-
- After successful completion of the job and approval of the Work Order.
 - ERP syncs WO# from AgriERP when work order gets in "In-progress" state.
 - Materials consumption and , resource and machine hours will be posted from AgriERP in the project item and expense journal respectively. F&SCM sends to AgriERP the success message if the journal is successfully posted, otherwise the failure message if the journal failed to post.
 - Steve will go to the Projects
 - Then click the desired Project
 - Open the Project Statement
 - This is how Steve can verify and validate the consumption and make the next steps. Alternatively he can also refer to standard F&SCM project reports for consumption.
-

Note: End of flow.

FarmApp

David | Farm Operation Manager

- David logs in to the AgriERP Web app with his credentials.
 - He navigates to the work order management page and clicks on the "Create New Work Orders" button.
 - A side window opens with the Work Order creation form.
 - David completes all the required fields.
 - He then submits the Work Order.
 - The Work Order is synchronized with the Mobile App.
-

- Once the Work Order is completed by a worker like Martin(MO) in our case
- David reviews the Work Order
- Adjust the consumption/Hours of resources and assets if required
- Mark the Work Order Complete by clicking on the "Mark As Done" button.
- He then writes the message in the pop-up and clicks Approve
- The Work Order moves to the Completed state
- All synced to the D365

Mobile App

Martin | Machine Operator

- Martin Login to the AgriERP App on his Tablet
- Tap on the Work Orders tab
- Navigate to the Work Order he needs to start the job
- Now he can start the job by
 - Going to the Plots tab
 - Slide left and Click Start Block
- OR
 - Clicking three horizontal dots above
 - A bottom sheet opens
 - Click the Start Work Order
- Then the Work Order starts and he starts working
- He can then end the job/Save Progress by
 - Sliding left the Work Order card OR via three horizontal dots
 - He can enter the progress.
 - He can mark the job complete by turning on the Mark Complete toggle
 - He may also add Additional Comments, if any
 - If he clicks Save Progress, the job will be paused
 - He may again start the job if it's not completed already.
 - If he clicks the End Job, the Work Order will be consumed 100%
 - The progress is synchronized with the Web app.
 - Resource hours are distributed if work is on multiple plots.

Business Use Cases

1. **Work Order Creation**

The Manager creates a Work Order in AgriERP Web App against the required project, activity, plot, resource, machine, and material details.

2. **Work Order Execution**

The Supervisor, Responsible Person, Machine Operator, or Farm Hand starts and completes the assigned Work Order job from the Mobile App.

3. **Resource and Machine Hour Capture**

The system captures resource and machine hours based on the actual job start and end time.

4. **Single Plot Hour Calculation**

If the resource and machine work on one plot only, the complete job duration is assigned to that plot.

5. **Multiple Plot Hour Distribution**

If the same resource and machine work on multiple plots during overlapping time, the system distributes hours based on plot acreage.

6. **Multiple Work Order Distribution**

If the same resource and machine are used across multiple Work Orders during overlapping time, the system recalculates and distributes hours across all impacted plots and Work Orders.

7. **Overlapping Entry Handling**

If a user creates an overlapping entry, the system prompts the user before voiding the existing entry and recalculating impacted hours.

8. **Hour Adjustment**

If job time, plot, resource, or machine is updated, the system recalculates hours and applies only the difference as an adjustment.

9. **Manager Review and Approval**

The Manager reviews resource hours, machine hours, material consumption, and adjustments before approving the Work Order.

10. **Posting to D365**

After approval, AgriERP syncs approved material, resource, and machine consumption to D365 for project costing.

11. **Paycom Variance Review**

Paycom actual payroll hours and cost are compared with AgriERP productive Work Order hours and cost to identify variance.

Validations:

1. Job end time must be greater than job start time.
2. The system must validate overlapping time for the same resource and machine combination.
3. If an overlapping entry already exists for the same resource, machine, plot, and Work Order, the system must show a confirmation prompt before voiding the existing entry.
4. If the user confirms, the existing entry will be voided and the new entry will be created.
5. After creating or updating an entry, the system must recalculate all impacted overlapping records across plots and Work Orders for the same resource and machine.
6. If only one plot is active during a time segment, the full segment hours must be assigned to that plot.
7. If multiple plots are active during the same time segment, hours must be distributed based on plot acreage.
8. The system must apply the formula: $\text{Weighted Hours} = \text{Overlap Hours} \times \text{Plot Acres} / \text{Total Active Acres}$.
9. Total distributed hours must not exceed the actual working duration of the same resource and machine.
10. If start time, end time, plot, resource, or machine is updated, the system must recalculate impacted records and calculate the adjustment impact.
11. $\text{Adjustment Impact} = \text{After Update Hours} - \text{Before Update Hours}$.
12. The system should maintain before-update hours, after-update hours, and adjustment impact for audit visibility.